Network Defense Intelligence Exercise

Independent Research and Adversarial Feasibility Assessment

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Status: Independent first-pass desk study for peer review. Not a tested architecture, procurement estimate, legal opinion, approved funding application, or ratified canon.

Distribution: Collaborative AI review and sanitized GitHub publication subject to owner review.

Executive decision summary

Recommendation: GO for a bounded feasibility and coverage study; REVISE the proposed architecture before a prototype; NO-GO for a national-scale live deployment claim. [R / D / high confidence in the need to narrow scope; moderate confidence in incremental defensive value.] This is a recommendation derived from the evidence, not a demonstrated project result.

1. Goal: protect a defined mission and genuine assets while optionally delaying adversaries and collecting useful intelligence. Capture is secondary to preventing loss. The exercise should test incremental protection, not count disappearing black lines. MITRE Engage supports goal-driven engagement rather than merely installing decoys. [S13]

2. Importance: a study is justified because meaningful deception and automation precedents coexist with demonstrated detection, coverage and scaling limitations. The strongest reason to run the study is to find where the concept should not be trusted, before it affects production. NIST frames cyber resilience around anticipating, withstanding, recovering and adapting, not proving invulnerability. [S02-S08, S12]

3. Cost: illustrative, bottom-up 2026 USD estimates are T1 desk study $59,400-$312,500 (base $138,000); T2 isolated prototype/range $191,400-$1,525,000 (base $600,300); T3 bounded pilot $693,000-$5,000,000 (base $1,973,400); T4 multi-organization research demonstration $1,870,000-$13,750,000 (base $5,336,000). These are alternative stage budgets, not quotes; arithmetic and exclusions appear in Appendix D. National operation is not responsibly estimable from the present inputs. [R / D / low confidence in amounts, high in arithmetic; S14 is wage context only.]

4. Funding: the best verified routes are a university/nonprofit research partnership under NSF SaTC RES or an eligible US small business under NSF SBIR/STTR. The November 4, 2026 SBIR deadline has passed; the official solicitation lists March 4, 2027 next. SaTC accepts proposals anytime with annual January/September target dates. Neither route establishes this project's eligibility or assures an award. [S15-S17]

5. Required attention: define scope, authorization, genuine-asset outcomes, eligible diversion points, false-positive safeguards, control-plane independence, operational owner and deployment acceptance criteria.

6. Bypass: the highest-priority threats are valid-access abuse, preexisting compromise, management/update-plane compromise, unmonitored third-party paths, and human/physical workflows outside the overlay. They do not require defeating the decoy. CISA's Volt Typhoon advisory documents legitimate-account abuse and persistence in some environments for at least five years. [S11]

7. Evolution: test repeated encounters and attacker non-engagement, not a scripted progression to total absorption. Disclosure need not destroy value: a controlled study of over 130 professional red teamers found the combined presence and explicit mention of deception had the greatest effect on measured attack behavior. This does not prove enduring deterrence. [S08]

8. Realism and modeling: virtual honeynets, dynamic generation, address mutation, protected observation, coordinated IoT deception and nation-like cyber exercises are real. This search did not establish the full national-scale, seamless, three-second, all-attacker combination. Absence from this review is not proof that no nonpublic version exists. [S02-S10]

9. Lessons: lightweight emulation scales but can be recognizable; high-interaction monitoring has assurance and overhead costs; simulation success can be built into the scoring rules. A major modeled precedent declares the attacker lost on exploiting a honeypot, which is precisely what this project must not assume. [S02-S06]

10. Changes: replace universality with bounded claims; define capture states; make cycling tunable; preserve ordinary access controls; test bypass and legitimate users; require signed, reviewable transfer and rollback. Keep creative canon amendments separate and pending ONESTAMP. [R,C / D / high; S01]

1. Execution method, independence and limits

This is the full desk-research execution of the prompt's workstreams, not execution of the proposed laboratory experiments. Public web search, page extraction, supplied-document inspection, project-file review, and Python cost/statistical calculations were used. No network probe, cyber-range trial, human-subject study, funder contact, vendor quote, or repository modification was performed. No peer research report was available for comparison. Peer prompts had already been reviewed and influenced v1.1; independence is therefore of the first research synthesis, not blindness to prompt ideas.

The review is targeted and reproducible at the query/topic level, not a systematic review with exhaustive database coverage. Searches favored public English-language primary evidence and official funding pages. Foundational papers were retained despite age because they directly expose the proposed architecture's tradeoffs. Search returned newer and commercial material, but no unsupported efficacy, price, or national-scale claim was imported. Access date for external sources is November 5, 2026. Appendix F records searches, retrieval depth, exclusions and unmet evidence needs.

Bias-control correction: hypotheses were not formally preregistered in a separate timestamped artifact before this session's first search. The hypotheses below organize the analysis retrospectively; they are not falsely described as preregistered. A future experiment must preregister before collecting outcomes.

H1: selective deception adds measurable benefit beyond matched baseline controls; disproof would be no mission-risk improvement at acceptable cost and harm. H2: rapid cycling improves that benefit; disproof would be no advantage over static/slower/randomized controls or excess instability. H3: persistent deception value survives attacker awareness; disproof would be collapse of benefit or displacement to unprotected assets. H4: incremental cost is fundable and operationally supportable; disproof would be recurring labor or assurance burden above the owner's threshold. None of these hypotheses was experimentally adjudicated here.

Finding labels: R real-world feasibility; C canon realism; N narrative design. A means primary/verifiable source, B credible secondary, C single/vendor commentary, D researcher inference, E speculation. Source grade and confidence are separate: an authentic source can have weak external validity. Findings cite S-IDs resolved in Appendix G; original tool citation tokens are not required to use the exported report. Costs and proposed acceptance criteria are D, not sourced facts. Qualitative likelihood rankings are planning judgments, not population probabilities.

2. Source dossier audit and project context

[R,C / D / high] The PDF is 25 pages; local parsing found 15 DOCX tables. Normalized word counts differ (6,358 PDF tokens and 6,283 DOCX tokens under the analysis tokenizer). Reading-order and formatting differences make a token-sequence score unsuitable as a semantic equivalence proof. The principal sequence, Q20, ONESTAMP language and sleeper-implant material occur in both. No material contradiction between formats was identified in the inspected core material, but a line-by-line editorial reconciliation remains outstanding. Treat them as two exports of one source, not independent corroboration. [S01]

AUD-01: pending approval versus locked canon. Severity high for governance, not a desk-research blocker. Treat locked as the dossier's asserted creative baseline, not independent ratification. Nathan decides canonical status. PDF pages 1, 3, 15, 25.

AUD-02: reviewer counts versus distribution/roles. Severity medium. Five named reviewing systems in the summary differ from the six-system recipient/role roster including Claude. Record actual artifacts and contributors; no assumption of six completed reviews. Pages 1-2, 13.

AUD-03: asset tiers. Severity high before specification. Five-tier wording, Tier 0 through Tier 5 examples, and Tier 0/Tier 1 critical-control labels conflict. Replace ad hoc numbers with a defined taxonomy and separate decoy fidelity from business criticality. Page 12.

AUD-04: incorrect cross-reference. Section 4.4 refers to Q17 for transfer; Q14 addresses transfer. Editorial correction, low severity. Pages 9 and 18.

AUD-05: scenario count. The roadmap says five goes-wrong scenarios, but Section 9.4 labels four; the five failure modes in 9.3 are a different list. Correct count and references, low severity. Pages 21-25.

AUD-06: timeline consensus. The summary's broad convergence should retain the table's different primary placements. Medium narrative severity. Pages 2 and 6.

AUD-07: capture conflation. High engineering severity. A line can vanish because of successful isolation, attacker departure, sensor failure, routing loss or dashboard error. Never make a disappearing line proof of safety. Pages 3, 8, 10-12.

AUD-08: protection-before-generation seam. High. If genuine access is already neutralized, deception adds intelligence or delay, not the original protection. If migration is needed to neutralize access, the ordered sequence must expose its timing and failure conditions. Page 3.

AUD-09: sole visibility and authority. High before operational design. The sole Architect is both a single operational dependency and a high-value target; separate safety/ground-truth observation from ordinary collaborator views. Page 13.

Project context: the UIARC preliminary charter explicitly calls itself unratified worldbuilding and disclaims real-world authority. Its useful conceptual distinction is admission versus use versus durable adaptation, but its universal-intake doctrine must not replace real-world privacy/collection limits. It is not a new authorization for this research. [C / A for what that draft states; S21]

3. Goal, necessity and alternatives

[R / D / moderate] Recommended objective: in a consenting organization's nonproduction enclave, determine whether selective deception and verified isolation reduce compromise of designated mission assets relative to existing controls, without increasing benign-user harm, analyst burden or total cost beyond agreed limits. Secondary objectives are eliciting behavior, shortening response, and discovering blind spots. Attribution support is not guaranteed attribution; deterrence is a separate measurable hypothesis.

Why study it? Honeyd, coordinated IoT deception and human-subject results justify a focused empirical question; fingerprinting and monitoring/scaling papers justify skepticism. Why not national scale now? The source supplies neither a national authority model nor a denominator of all attackers, per-protocol handoff mechanism or resource envelope. Those missing requirements cannot be solved with additional narrative certainty. [S01-S08]

Use "bounded research exercise" now, "prototype" after G2, and "pilot" only after operator consent and independent review. A live national absorption operation is not an exercise merely because observers watch it. Sponsor, control team, injects, evaluators, ground truth, hot wash and improvement plan should be explicit.

Counterfactual: spend the same money on identity hygiene, supported edge devices, central logs, network segmentation, recovery rehearsals and incident response. CISA prioritizes patches, phishing-resistant MFA, logging and lifecycle management in the relevant threat context. Deception must demonstrate additional value rather than borrow the baseline's security effect. [S11]

An equal-budget T2 comparison should give the conventional arm the same engineering hours and infrastructure ceiling as the deception arm, with independent scoring. A cheap static-token arm may deliver a useful alert at much lower fidelity cost; that would be a valid finding, not failure to implement enough AI. No funding means loss of a learning opportunity, not automatic national vulnerability; conventional safeguards remain valuable.

Minimum research product: a validated coverage map and one well-controlled negative or positive result. Minimum deployable capability: one authorized entry path, explicit supported protocols, independent isolation verification and a documented noncovered-path policy. Research novelty must exceed assembling existing tools; a generalizable method for measuring deception under bypass is a more credible contribution than a new branded honeypot. [D; S06, S13, S16]

4. Capability and architecture assessment

[R / D / moderate] The architecture should have separate genuine-asset protection, engagement, management, and evidence/safety planes. This is a proposed separation of responsibilities, not proof of a seamless translation layer. Genuine-asset access controls must work if the decoys are absent, overloaded or recognized. Preserve clear provenance distinguishing synthetic records from evidence; give independent evaluators read-only access to ground truth outside the deception controller.

Detection and classification: demonstrated for selected observable behaviors; unsupported as literal purpose-reading. Mapping: partially demonstrated for observed attack paths, not all upstream proxies, identities or latent footholds. False services/topologies: demonstrated at smaller scale. New-session diversion under administrative control: demonstrated in virtual honeypot frameworks. Transparent migration of arbitrary established encrypted application state: unsupported at the claimed generality in this review. Verified isolation: a feasible bounded requirement, not a guarantee over unknown paths. Dynamic address mutation: research-stage/model-and-prototype evidence, not a universal three-second optimum. Universal capture/national security closure: unsupported. Packaging/update transfer: a conventional engineering deliverable in principle, but no NDIE implementation, inspection or acceptance test exists. Appendix A makes statuses explicit. [S02-S08]

TLS/application identity, authenticated sessions, application transactions, client caches and external validation constrain what can be changed unnoticed. Redirecting a new connection to an already controlled service is not transferring an existing compromised endpoint or its authority. Encryption does not make redirection impossible in every deployment, but the defender must state the legitimate termination, trust and state-control assumptions. Rotating only decoy identifiers does not rotate every attacker-held secret or revoke real credentials. [D; S02,S06]

Three-second assessment: no retrieved study establishes three seconds as an optimum for this concept. Reti et al. use abstract action steps, not wall-clock seconds. A UI pulse, address mutation, TLS key update, server reboot and database state change are different actuations. Treat them separately, compare 3/30/300-second laboratory schedules plus event-driven/randomized/static arms, and never force a production protocol to match a story rhythm. Numerical schedules here are proposed test settings, not attack instructions. [S06; D]

The phrase "quantum-inspired" specifies neither algorithm, complexity, measured speedup nor needed hardware in the dossier. Its contribution is therefore unsupported until an explicit classical baseline and reproducible benchmark exist. Generative AI might synthesize decoy text; it is not necessary for route control or session-isolation proof. Keep attacker-derived content outside trusted prompts, preserve immutable policy, and gate all learned changes through review. [D; S01]

Scale depends on concurrent interactions and state complexity, not the number of advertised addresses. A low-resolution address facade and a high-fidelity live national infrastructure are not comparable units. Potemkin-related scaling reported in Bailey et al. is secondary within that paper; this review does not use those IP-per-host figures as modern capacity estimates. [S03]

5. Evidence from comparable work: what worked and what did not

[R / A for reported observations; moderate external validity] Ten relevant cases are summarized in Appendix B. They are not ten replications of NDIE. Three independent primary studies provide adverse findings: Honeyd reports distinguishable state behavior and performance limits; Bitter Harvest finds systematic fingerprinting and stale deployments; VMM sensors show monitoring/TCB tradeoffs and benign administrative false positives. Bailey et al. adds direct evidence that unscanned paths escape darknet observation. These are limitations, not evidence that all deception fails. [S02-S05]

Honeyd's 2004 evaluation made selected virtual network personalities convincing to Nmap at the time and supported thousands of lightweight honeypots on contemporary hardware. The paper itself identifies differences in untested TCP behavior and coupled performance that can reveal emulation. Lesson: demonstrate fidelity against independent verification, not only the same tool used to design the facade. [S02]

Bailey et al. demonstrate filtration and distributed observation across 30 organizations, but explicitly note that threats avoiding unused blocks are not observed. Their high-fidelity analysis needs careful filtering; source overlap does not automatically create broad scaling savings. Lesson: "the threat did not appear in our trap" is not "the threat did not exist." [S03]

The VMM study observed five separate compromises over three months and tested 46 additional collected exploits. A small sensor set was useful for that population, but the authors expressly disclaim complete capture. Their benign-use experiment produced approximately one false positive per week from administrative actions, and faster monitoring could enlarge the trusted computing base. Lesson: separate evidence custody, false-positive adjudication and assurance cost. [S04]

Bitter Harvest identified 7,605 honeypot instances across nine implementations and reported equal error rate 0.0183 in its studied fingerprint classifier. That is not a universal present-day honeypot detection probability. Maintenance data also contradict the idea of deploy-and-forget realism. Lesson: design and continuously test for detectability; do not equate randomized banners with credible underlying behavior. [S05]

Reti et al. model honeypots and address mutation over 10/50-host scenarios with three seeds and 100 repetitions per parameter combination. The model ends the attacker run when a honeypot is exploited, omits credentials in the reported setup and lets address mutation erase agent knowledge of previously exploited hosts. These assumptions can manufacture containment that a real system must actually enforce. One-host loss is also a different endpoint from compromise of all sensitive hosts. Lesson: let attackers continue, retain knowledge, avoid decoys and succeed via independent paths. The authors warn results are specific to simulation parameters; the text's interval/result descriptions also warrant checking figures before quoting exact effects. No effect percentages are imported into NDIE. [S06]

Guan and Cao's CPDS coordinates IoT honeypots to preserve cross-device cyber-physical consistency using dependency rules; online deployment, human evaluation and simulation are reported. It is evidence against the blanket assertion that physical dependencies make deception impossible. It is not evidence that every independent physical measurement can be simulated. Classifier error percentages from the linked paper are not reused as NDIE detection precision because the experimental meaning differs. [S07]

Ferguson-Walter et al. compare four conditions among over 130 professional red teamers and report greatest measured behavioral effect when decoys and explicit information about deception coexist. Lesson: secrecy versus deterrence is an empirical design choice; avoid assuming awareness immediately defeats deception. Multi-year campaign effectiveness remains unknown. [S08]

Locked Shields demonstrates nation-like exercise construction, not control over a real nation's networks. Official 2025 reporting describes 8,000 virtual systems; 2026 reporting describes 16 teams, more than 4,000 participants from 41 nations, and roughly 8,000 attacks over two days. Systems and attacks are different counts, and the years must not be blended. Public releases do not disclose a controlled deception treatment effect or detailed failure postmortem; none is invented here. [S09]

DARPA CGC demonstrates automated vulnerability reasoning/patching in a purpose-built air-gapped tournament with seven finalists. Its prizes are not the cost of a national defense architecture. The transferable lesson is bounded automation with functional scoring, not guaranteed automatic countermeasure synthesis for arbitrary attacker behavior. [S10]

CISA's Volt Typhoon advisory shows why already-trusted access, long dwell and weak management boundaries belong in the exercise. It is evidence of relevant adversary behavior, not a test in which NDIE was deployed and defeated. [S11]

Answer on exact precedent: this targeted search found component and combined honeypot/MTD modeling, but not the full dossier architecture at national operational scope. National-looking exercises validate logistics and scenario realism only within their constructed boundaries. Nonpublic deployments and inaccessible studies limit certainty.

6. Bypass-first adversarial assessment

[R / D / high confidence in coverage logic; likelihoods provisional] The invariant is simple: a threat that does not cross a defender-controlled decision point cannot be contained merely by changing what that point presents. Decoy avoidance and decoy defeat are not the same thing. Appendix C supplies assumptions, warnings, independent controls, safe tests and residual risk for ten classes.

Provisional top five, ordered for a typical enterprise/critical-infrastructure-adjacent enclave rather than a national probability distribution:

1. Valid-account and trusted workflow abuse. CISA documents legitimate-account use. Monitor identity and transaction authorization, not just inbound exploit signatures. A user may reach real data without ever selecting a decoy. [S11]

2. Compromise already inside before activation. Existing credentials, background jobs and dormant footholds bypass the premise of a newly arriving adversary. Require threat hunting and recovery assumptions; do not count silence as eviction. [S11]

3. Management, supply/update or orchestrator compromise. An attacker who changes routing policy, identity rules, logs or deployment packages can invalidate the evidence that isolation succeeded. Treat this as a shared dependency with independent safety checks. It is an architectural risk inference, not an observed NDIE incident. [D; S04,S11]

4. Third-party/SaaS and unmanaged alternative paths. If the owner lacks coverage or control, a decoy cannot repair the missing authority. Catalog dependencies and put conventional authorization at the real asset. [D; S03,S11]

5. Insider, maintenance and physical-process paths. Actions originating inside the real trusted environment may have no migratable remote session. Preserve local safety functions and human separation of duties. Exact prevalence depends on the asset and remains unmeasured. [D; S07]

Additional scenarios: resource exhaustion that forces engagement shutdown; deliberate telemetry poisoning; short destructive actions completed before detection; innocent shared infrastructure mistaken for a culprit; and an adversary who knows the real asset through an independent channel and refuses attractive decoys. No numeric scan recipes or exploitation steps are needed to test these classes safely.

Which make universal security untenable? Any single in-scope real-asset path that lies outside effective prevention, any preexisting authority not revoked, or any compromised protection/evidence plane is enough to defeat a claim that all real assets are fully secured. This does not imply that a scoped deception program has no useful effect; it means its success claim must be narrower.

Capture state machine: suspected -> observed -> diverted -> isolation independently verified -> evidence retained -> closed for the defined scope/window. Separate terminal states are attacker departure, visibility lost, unresolved, false-positive released and safety abort. Only the independent verifier may mark isolation verified. Count sessions, identities, workloads and campaigns separately; never infer people from source IP counts. [D; S03,S04]

7. Evolution, adaptation and uncertainty

[R / D / moderate] Generation 0 is unaware; G1 knows deception exists; G2 recognizes broad characteristics; G3 checks authenticity independently; G4 attempts defender misclassification; G5 assumes both sides may deceive. These are test cohorts, not a universal historical sequence or timed forecast.

Hours to days: protection and evidence integrity dominate; freeze learning, establish benign baselines, retain independent snapshots and record departures. Weeks to months: repair stale personas and recurring false positives, audit bypass routes, measure analyst effort and update operator playbooks. One to three years: assess disclosure, staff turnover, protocol changes, model drift, supply-chain changes and re-entry through new integrations. Five to ten years: reauthorize, retire unsupported components, reassess vendor lock-in and mission relevance; do not extrapolate today's attack rates. [D; S05,S08,S12,S13]

Best trajectory: static or modestly adaptive decoys expose otherwise invisible movement at acceptable workload; knowledge of deception still slows some actors. Expected uncertainty: some actors engage, others avoid or shift paths; fidelity maintenance and analysis consume more effort than assumed. Worst trajectory: false confidence displaces baseline protection, attackers poison updates, and the partner deploys a brittle artifact without rollback. These are scenarios, not probabilities.

Proposed reassessment: monthly during a prototype, quarterly during a consenting pilot, and after material identity, vendor or topology changes. Track mission compromise, bypass coverage, benign misrouting, time to independent verification, stale-decoy fraction, cost per adjudicated useful event, and manual recovery duration. Pause automated changes if any safety metric worsens beyond a preregistered envelope.

Do not equate longer dwell in a decoy with benefit if real assets are simultaneously compromised. An attacker can sacrifice one session to distract analysts. Similarly, a decreasing alert count may be successful deterrence, detector failure or migration to another path. Require other evidence before interpreting it. [D; S03,S08]

8. Experimental program and acceptance rules

[R / D / proposed, not performed] Design one nonproduction enclave of 20-50 logical hosts, one mission application and synthetic identities/data, with protected and decoy paths, an independent evidence collector, isolated management, and no route to actual production. These sizes are planning assumptions, not minimum technical requirements. Begin deterministic; add generative components only if a concrete fidelity limitation justifies them.

Arms: A matched baseline controls; B baseline plus static deception; C baseline plus adaptive deception without rapid cycling; D baseline plus candidate cycling policies. Maintain equal resource/time ceilings and independently score genuine mission compromise. Include real-asset paths that never encounter a decoy and scenarios where the attacker abandons or returns. Do not stop an attacker merely for touching a honeypot.

EXP-01: incremental protection. Null: no improvement over A. Randomize equivalent scenario assignments and operator order; primary endpoint unauthorized synthetic mission-data access within a fixed session window. Secondary analyst effort and time to independent isolation. Separate prevention benefit from intelligence yield.

EXP-02: bypass coverage. Null: decoy engagement does not predict coverage of nonengaging threats. Use ten abstract path classes from Appendix C with evaluator-known ground truth. Outcome: exposure of protected test assets despite visually successful decoy containment; report coverage denominator and missing paths.

EXP-03: cycling. Null: fixed three seconds has no advantage over static, slower, randomized or event-triggered mutation. Vary one actuation layer at a time. Record established-session continuity, independent reachability, benign p95 latency and forensic reconstructability. A UI pulse is not an independent treatment.

EXP-04: repeated learning/disclosure. Null: benefit does not persist across encounters. Use separate trained/untrained cohorts, balance experience, randomize disclosure and decoy presence, and retain attacker memory across episodes. Treat people as clusters; do not count their sessions as independent subjects.

EXP-05: benign collateral effects. Null: diversion exceeds the acceptable false-positive burden. Replay synthetic admin, monitoring, partner and emergency-maintenance workflows, including unusual but authorized behavior. Record false diversion per benign session and reversibility; independently adjudicate all positives.

EXP-06: failure and recovery. Null: isolation and evidence cannot survive controller/sensor/update failure. Inject benign test faults and unsigned test-update rejection; verify protective controls remain effective with engagement disabled. Measure time to abort/recover, routing integrity and evidence gaps. No real malicious payload or third-party contact is required.

EXP-07 optional: poisoning/drift. Feed labeled synthetic misleading records into an offline candidate learner; verify none alters trusted policy absent approval. EXP-08 optional: federation with two consenting test owners and mutually independent controls; measure authority handoffs, disclosure and rollback.

Sampling: start with 20 pilot trials per machine scenario to estimate variance, not to declare efficacy; then power the primary comparison. Illustrative independent-binomial calculation for a drop from 40% to 20%, alpha .05 two-sided and 80% power yields about 82 trials per arm; clustered humans, multiple comparisons and repeated episodes require more or a different design. These baseline rates are assumed, not observed NDIE rates. Report confidence intervals, preregister the primary endpoint and label secondary exploration. Use holdout topologies/seeds, version images and policies, and retain all failed runs.

Candidate go criteria, requiring owner ratification before testing: meaningful mission-risk reduction (example 20 percentage points in the assumed scenario), no isolation escape, no production route, p95 benign latency increase below 5% on the chosen benchmark, and manageable false diversion. The 5% and 20-point thresholds are proposed values, not standards. Any unauthorized external effect aborts immediately regardless of statistics. A significant but tiny engagement gain with no mission benefit is insufficient.

False-positive planning example: with attack prevalence 0.1%, sensitivity 95% and false-positive rate 0.1%, expected alert precision is only about 48.7%. Those are hypothetical inputs to illustrate base-rate risk, not detector measurements. Roughly 3,000 independent benign trials with zero false diversions only bound the 95% rate near 0.1% under the rule-of-three approximation; independence and representativeness matter.

9. Economic feasibility and opportunity cost

[R / D / low-to-moderate confidence] The figures in the executive summary are constructed planning scenarios, not empirical bids. The base monthly loaded labor allowance is $18,000 per full-time-equivalent person-month; low/high use $12,000/$25,000 with different staffing totals. BLS reports $129,180 May 2025 median annual security-analyst wages; dividing by 12 gives $10,765 wage-only per month. The base allowance is about 1.67 times that benchmark to allow benefits, seniority mix and organizational support. That multiplier is an assumption, not a BLS burden rate. [S14]

Loaded labor includes salary, employer benefits/payroll and routine organizational support. Do not add a second blanket overhead rate. A university must replace this assumption with negotiated indirect rates and a role-based payroll budget. Specialist external assessor/counsel fees appear separately, not inside core person-months. This study has no binding cloud, range, insurance or hardware quotes; nonlabor values are allowances. Region: US remote/hybrid team without clearance-required facilities or nationwide 24/7 coverage.

Base staffing: T1 six person-months over three months (research lead 2, analyst 2, architect 1, program/documentation 1). T2 24 person-months over six months (lead 3, engineering 12, evidence/data 3, test analyst 4, program 2). T3 72 over 12 months (lead 6, engineering/integration 30, operations 18, evaluation/data 12, program 6). T4 180 over 18 months (lead 12, engineering 72, operations 48, evidence/validation 30, program 18). These are average staffing equivalents, not hiring commitments. Low/high totals are in Appendix D.

The five biggest drivers are skilled labor, fidelity/integration complexity, independent assurance, telemetry retention/analysis, and sustained operator coverage. Address count alone is a poor cost driver. AI inference is a cost only if included; it is not necessary for a first experiment.

Base nonlabor allowances (USD thousands): T1 compute/storage 1, tools 1, external review 6, stakeholder workshops 2, release/document preparation 2 = 12. T2 compute/network 18, storage 6, tools 6, range/facilities 12, independent assessment 24, counsel/privacy 12, training/release 12 = 90. T3 compute/network 48, storage 24, tools 24, facilities/OT emulation 90, independent assessment 84, counsel/privacy 36, operator training/partner integration expenses 72, travel 12, insurance allowance 30 = 420. T4 corresponding categories 180, 90, 90, 300, 240, 120, 240, 60, 80 = 1,400. Vendor implementation labor belongs in core staffing unless expressly subcontracted; do not count it twice.

T3 base recurring limited operations: 24 person-months/year at $18,000 plus $120,000 recurring nonlabor and 15% reserve = $634,800/year. Three operating years after T3 setup: about $3.878 million total; five: $5.147 million, constant 2026 dollars, no inflation/discounting. This is a limited program, not 24/7 national service. T4 illustrative recurring base: 60 person-months plus $400,000 nonlabor and 15% reserve = $1.702 million/year; three/five post-setup operating years give $10.442m/$13.846m. These are scenario totals, not validated lifetime cost.

Sensitivity: in T2 base, 25% more labor adds $124,200 after reserve; doubling the $24,000 compute-plus-storage allowance adds $27,600. In T3, doubling $24,000 annual modeled storage allowance adds $27,600; 12 additional operating person-months add $248,400. National staffing, retention, procurement and mission safety could dominate far more; T5 is non-estimable until ownership, supported services, concurrency, SLOs and authority are specified.

Equal-budget alternative T2 base: 24 person-months and the same $90,000 nonlabor can be allocated to identity/segmentation/recovery testing instead of deception. No evidence here establishes which yields more avoided loss for the user's organization. Break-even requires incremental risk reduction times credible consequence value to exceed incremental lifetime cost; use scenario losses, not a borrowed universal breach average. Cheapest discriminating next work is a 2-4 person-week boundary/tabletop exercise with one asset owner before committing the whole T1 program. At an assumed $100-$150/hour, 80-160 hours is $8,000-$24,000 labor only. Existing volunteer/school resources may lower cash outlay but not erase opportunity cost.

10. Funding pathways and readiness

[R / A for program terms, D for fit] Verified option one: NSF SaTC 2.0 RES, NSF 25-515. Up to $1.2 million total over up to four years, not $1.2 million annually. Eligible submitters include qualifying US higher-education institutions and US nonprofit research/education organizations. The PI eligibility rules require an appropriate full-time paid research/teaching or tenure position at an eligible institution; an independent concept owner is not automatically eligible. The proposal should study generalizable coverage, uncertainty and human/technical trust, not purchase national infrastructure. Proposals are accepted anytime, with last-Monday January/September targets; check updates before submission. Voluntary committed cost sharing is prohibited. T1/T2-sized research fits more plausibly than the T3 base pilot. Institutional registrations, data management, applicable human-subject review, IP/publication and security terms require research-office approval. There is no established team or university commitment. [S16]

Verified option two: NSF SBIR/STTR, NSF 26-510. Phase I up to $305,000 for 6-18 months; Phase II up to $1.25 million, typically 24 months, for eligible NSF Phase I awardees. Requires a qualifying small business, Project Pitch invitation for Phase I, and appropriate PI employment (normally at least 51% with the company). STTR additionally needs a nonprofit research-institution partner. No PhD is required by the displayed PI rules. The company/IP ownership language is not freedom from all award obligations. Voluntary committed cost sharing is prohibited under the main solicitation; supplements can have separate conditions. Active SAM and institutional registrations are needed. [S15,S17]

Deadline: yesterday's November 4, 2026 window is closed as of this report. March 4, 2027 appears next in the official notice. Do not rush a proposal based on a third-party "active" badge. Fast-Track has special invitation/timing rules and a displayed discrepancy: the summary says $1,555,000, while a body sentence says $1,555,555, although the $400,000 and $1,155,000 components sum to $1,555,000. Seek program-officer clarification; this report does not build a funding plan around the conflicting extra $555. [S15]

Conditional routes, not verified open opportunities: a university internal seed award; operator-sponsored study; industry research contract; in-kind cyber-range access/cloud; DOE/national-laboratory collaboration; DARPA/DoD research; state critical-infrastructure partnership; philanthropic or international consortium. No active matching solicitation, award ceiling or eligibility determination was verified for these routes. Appendix E lists the missing checks instead of invented amounts.

DARPA warning: the official historical I2O BAA retrieved closes in 2025. 2026 aggregator results conflict on office name, pause status, invitation rules and deadlines. No current DARPA opening or typical award amount is certified here. Recheck the actual amended federal notice before any application. This limitation is a finding, not an excuse to reuse a stale listing. [S19]

First-dollar sequence: settle owner and scientific question -> obtain university or eligible-company partner -> budget a narrow coverage study -> discuss fit with the official program contact -> prepare scoped concept note -> apply only after eligibility, terms, dates and security checks. Academic route emphasizes reproducible findings; company route also needs an unmet commercial problem, customer discovery, defensible technical risk and commercialization plan. Venture capital is investment with control/return expectations, not a research

grant. No personal eligibility facts are inferred about Nathan.

Concept note outline: mission/problem; prior art and gap; hypothesis including null; consenting test boundary; bypass-inclusive protocol; metrics and independent evaluator; privacy/safety case; scoped personnel/nonlabor budget; milestones and stop gates; intended outputs and IP/publication plan. Fund the question first, not a promise of universal capture.

11. Governance, safety and human factors

[R / A,D / high on need for review] US unauthorized-access and interception statutes are relevant issue areas; legal permission cannot be inferred from owning a decoy, reading an advisory, classifying traffic as hostile or calling work research. Cornell's displayed texts of 18 USC 1030 and 2511 contain prohibitions and exceptions that require fact-specific analysis. Provider/consent/trespasser exceptions, state privacy law, stored communications, cross-border data and government constitutional obligations are not resolved by this study. Obtain qualified counsel before real traffic or real user involvement. [S18]

Separate approvals for acquisition, retention, access, correlation, operational use, dissemination and learning. The UIARC draft helps illustrate separation but is fictional, not binding law, and does not justify universal collection. Set minimum necessary data and deletion schedules before any collection. Proposed lab policy: synthetic records only; pilot raw-content collection off by default until counsel and owner authorize it; no copying production credentials into decoys. [D; S21]

Safety case claim: disabling or compromising engagement must not authorize real-asset access. Evidence required: default-deny test routes, independent reachability tests, egress restrictions, immutable off-controller logs, signed artifact validation, bad-update rejection, break-glass operator access and tested recovery. The safety lead—not only the orchestrator—can abort. Unexplained evidence loss is an unresolved condition, not a green status. Controls must be tested after failure injection before a pilot.

No hack-back, external endpoint seizure or unauthorized tracing into third-party computers. Capturing a remote interaction is not taking control of the remote person's device. Maintain an incident contact and coordinate appropriate disclosure rather than publicly accusing IP owners. Shared infrastructure may be innocent. Permit release/escalation for benign misclassification and log the rationale.

Human-subject research: professional red teamers and ordinary users may require institutional ethics/IRB determination, consent, compensation and separation from employment evaluation. This study does not determine exemption. Budget counsel and ethics support rather than presuming a cyber range removes all human risk.

Transfer: deliver scoped interfaces, software bill of materials, provenance, policy/configuration manifest, signing and verification, supported-version matrix, receiving-operator training, license/IP terms, rollback and revocation. Do not auto-promote attacker-derived signatures or snapshots into trusted policy. A tethered black box may create vendor dependency and outage risk; source-code delivery may create disclosure risk. Compare them rather than deciding for narrative convenience. [D; S01,S04,S13]

Missing threats from stakeholder review: defenders—unknown pathways; operators—maintenance that looks suspicious; counsel—unauthorized third-party content; privacy reviewer—synthetic logs containing copied personal information; asset owner—safety dependency; red team—nonengagement; funder—unoriginal tool assembly; executive—misleading green dashboard; researcher—selection bias in who enters a trap. Repeat after each material change.

12. Required design changes and canon translation

[R,C / D / high unless noted] CHG-01 modify universal capture/security closure into a bounded observation-window claim with an explicit unknown category. CHG-02 modify purpose-reading into confidence-weighted evidence. CHG-03 modify complete national replication into staged, permissioned, selective emulation. CHG-04 modify fixed cycling into distinct visualization and actuation parameters. CHG-05 add independent isolation/evidence verification and conventional protections. CHG-06 modify transfer into an inspected signed release with receiving-owner acceptance. CHG-07 add preexisting compromise, valid identities, alternative workflows and nonengaging attackers to mandatory tests. CHG-08 add poisoning quarantine and review gates. CHG-09 correct tier taxonomy and audit inconsistencies. CHG-10 retain the black-line visual only with distinct unresolved/lost-visibility states. None is automatically an approved canon edit. Appendix H assigns owners and blockers.

[C,N / D,E / moderate narrative usefulness] Retain the national-scope image, three-second pulse and transfer beat as creative anchors if Nathan chooses, while narrating boundaries and cost honestly. A credible failure is an apparently successful external containment alongside an unrelated trusted maintenance path reaching the real asset. The Architect's achievement then remains meaningful but not universal; the cost is admitting that a beautiful model missed the owner/operator's real dependency. This is a proposed story, not an observed event.

[C,N / E / low engineering relevance] Recursive counter-simulation can remain a later Time Wars hook. Cyber research does not validate fabricated timelines, minds or civilizations. The more useful bridge is epistemic: deciding which observations deserve action when both sides can generate plausible records. Give bystanders agency and make corroboration costly. Do not require real collateral harm or a sacrificial ingress window merely to create tension.

The original OPEN questions about timeline placement, agency identity and a canonical name do not block desk research. Real operational owner, authorization, genuine asset boundary, learning permissions and receiving-party capability do block live work. Nathan alone ratifies canon; a real asset owner and qualified oversight must independently authorize operations.

13. Ranked questions, objections and action plan

QST-01 (blocking before prototype): what single mission/asset and consenting owner anchor the study? Nathan plus owner; completion is a signed boundary statement.

QST-02 (blocking before traffic): are all paths, identities and third parties authorized? Owner/counsel; completion is approved rules of engagement and data plan.

QST-03 (blocking before evaluation): what level of benign harm and mission compromise is unacceptable? Mission owner/evaluator; preregister endpoints and abort criteria.

QST-04 (blocking before funds): eligible university/nonprofit or company lead, available staff and IP goals? Nathan/research office; eligibility and contribution agreements.

QST-05 (blocking before build): which protocols/session phases can the defender actually control? Architect/engineer; capability test inventory, not universal language.

QST-06 (blocking before pilot): who independently verifies isolation and can halt? Safety lead; tested authority and recovery.

QST-07 (cost refinement): data rate, retention, concurrent high-fidelity sessions, operator availability and support budget? Engineer/operator; measured capacity worksheet.

QST-08 (canon only): event name, jurisdiction, placement and cost for Nathan? Nathan; ONESTAMP or unresolved status.

QST-09 (publication): public/private split, license, contributor permissions, retention and disclosure contact? Nathan/counsel; sanitized release checklist.

QST-10 (scientific novelty): what finding would be useful beyond existing products and models? Research lead; a falsifiable gap statement.

Strongest argument against funding: the concept currently aggregates established techniques with untestable national absolutes and may reward visible decoy activity over protected missions. Strongest argument for redesign: the protection boundary and independent verification, not a full synthetic country, are the actual difficult research problem. Strongest case for limited funding: controlled negative and positive results on bypass-inclusive evaluation could prevent expensive operational overclaiming and be publishable/reusable. [D; S02-S08,S13,S16]

Cheapest discriminating exercise: map one real workflow (without sensitive publication) and ask an authorized evaluator to identify an abstract genuine-asset route outside the proposed redirector. If such a path exists and conventional controls do not cover it, the universal design claim fails before a large simulation is built. This does not prove all scoped variants fail.

Now / days 1-30: research lead and owner reconcile audit, choose one mission, appoint safety/evaluation roles and draft pre-registration; estimate 2-4 person-weeks plus owner input. Deliver signed scope, ten path classes and decision log. No engineering gate is passed merely by having this report.

Next / days 31-60, conditional on G0-G1: engineer/evaluator specify matched arms, source-backed requirements, budget quotes and safe range acceptance; estimate 4-8 person-weeks. Obtain institution/company eligibility and counsel/ethics decisions. Stop if no generalizable question or consenting boundary exists.

Later / days 61-90, only after G2 authorization: build a minimal isolated demonstration and pilot measurement protocol, not a production pilot. Estimate 8-12 person-weeks depending on reused range capacity. Report failure runs and holdout results; scale timeline may exceed 90 days. Seek external funding only for the next justified stage.

Gates: G0 remains incomplete (owner/boundary/metrics unknown). G1 is conditionally supported for limited inquiry by precedents, not for full architecture. G2-G4 are not passed. Stop or substantially redesign for no advantage over matched baseline, unacceptable benign diversion, any isolation/production-routing failure, unverifiable evidence, lack of authority, unaffordable sustainment or an irreproducible apparent effect.

Readiness: suitable for shared peer critique and a proposed dossier change register; not ready for automatic canon ratification, operational specification or national budget request.

14. Pre-mortem, dissent and limits of the recommendation

Pre-mortem (scenario, not forecast): three years later the program has a polished dashboard but no independent coverage evidence; cloud and integration costs outpace staffing; decoy traffic is mostly irrelevant to the mission; a trusted update bypasses controls; an innocent operator is misclassified; the receiving agency cannot recover without the original Architect. The causes are misplaced endpoints, missing stakeholders and untested dependency assumptions rather than merely insufficient AI capability.

Dissent to the recommendation: mature commercial deployments or nonpublic state programs may already answer some questions better than public literature, so a new generic study could duplicate work. Conversely, legitimate-asset protection might be enough without engagement, making even a limited deception project unnecessary for some owners. We have not inspected products, surveyed owners or conducted meta-analysis, so comparative ROI remains unresolved.

What would change the conclusion? Independently replicated real-asset protection under nonengagement and valid-access scenarios; safe stateful handoff for specified protocols; sustained measured advantage after disclosure; defensible operator cost; or credible failure/maintenance evidence showing static alternatives dominate. A validated bounded result could justify expansion, not a leap to universal security.

Scope limitations: no experiments run; no current exhaustive national inventory or authority survey; no vendor price quotes; no original Potemkin paper full-text assessment; no detailed public Locked Shields failure postmortem; no independent replication of source results; no verified matching DOE/DARPA/state/private award open for this project; no complete jurisdiction-specific legal analysis. Three primary adverse studies

were found, but they are not three abandoned programs or universal refutations. These gaps are explicit research backlog items, not fabricated answers.

Report self-review: all ten user questions are answered or explicitly bounded; illustrative budgets and performance targets are labeled; the largest technical claims are not inferred from peer AI confidence; no peer-supplied national cost/capture numbers are used; no GitHub upload, outside contact or testing was performed.

Appendix A. Capability matrix and claim ledger

ID / capability	Status	Evidence and limit
CLM-P-001 / Observable intrusion classification	Demonstrated at smaller scale	R/A,D; S02,S04,S11. Accuracy population-specific; not intent reading.
CLM-P-002 / End-to-end path/campaign map	Partially demonstrated	R/D; S11. Observed path does not reveal every foothold or human.
CLM-P-003 / Genuine-asset isolation	Partially demonstrated	R/D; S12. Test boundary and failure assumptions required.
CLM-P-004 / Virtual false services/topology	Demonstrated at smaller scale	R/A; S02. Network facade not full national semantics.
CLM-P-005 / Dynamic high-fidelity generation	Demonstrated at smaller scale	R/A,B; S03,S04. Potemkin capacity only indirectly reviewed.
CLM-P-006 / New connection diversion	Demonstrated at smaller scale	R/A; S02. Requires owned routing/service boundary.
CLM-P-007 / Arbitrary established encrypted-session migration	Unsupported	R/D; search S02,S06,S07 does not establish general case.
CLM-P-008 / Address mutation plus honeypots	Research-stage	R/A; S06,S20. Models not production guarantees.
CLM-P-009 / Three-second optimal interval	Unsupported	R/D; S06 uses abstract action steps.
CLM-P-010 / Coordinated physical/device consistency	Demonstrated at smaller scale	R/A; S07. Dependencies bounded to modeled devices.
CLM-P-011 / Universal capture, no remaining threat	Unsupported	R/D; S03-S06,S11. Unknown denominator and bypass.
CLM-P-012 / Automatic policy updates from captures	Research-stage	R/D; no safe NDIE implementation validated.
CLM-P-013 / Temporal/civilizational containment	Speculative	C,N/E; S01. Not a cybersecurity finding.
CLM-P-014 / Value despite attacker awareness	Partially demonstrated	R/A; S08 controlled human study; long-run effect unknown.
CLM-P-015 / Goal-led planning and lifecycle resilience	Demonstrated	R/A; S12,S13 demonstrate published guidance, not effectiveness.
CLM-P-016 / NSF small-business route active	Demonstrated	R/A; S15. Eligibility and invitation remain unestablished.
CLM-P-017 / SaTC institutional route active	Demonstrated	R/A; S16. Appropriate eligible PI/lead required.
CLM-P-018 / NDIE tier costs	Unsupported as market prices	R/D; calculated assumptions, S14 context. Low confidence.

Counterevidence mapping: CLM-P-004/005 are qualified by S05 fingerprinting and S03 workload constraints; CLM-P-008/009 by S06's nonphysical steps and honeypot-terminates-run rule; CLM-P-011 directly conflicts with nonobserved paths in S03 and durable legitimate access in S11. Claim types and source locations are in the source ledger. No independent effect size is assigned where the underlying study was not fully reanalyzed.

Appendix B. Comparable-case register

Case	Study setting	Result / lesson / limitation
PRE-01 Honeyd (2004), S02	Prototype: thousands of lightweight hosts; fingerprint/personality tests	Worked: virtual services/routing. Adverse: unmodeled TCP behavior, shared performance, historic capacity. Transfer: selective facade, not universal fidelity.
PRE-02 Distributed darknet/honeyfarm (2005), S03	Field: 30 organizations; over 60 monitors in multi-year source data	Worked: reduce repetitive traffic before deep analysis. Adverse: nonscanning/avoidant threats invisible; aggregation not automatically cheap. Independent coverage negative.
PRE-03 VMM sensors (2006), S04	Prototype/field: two exposed guests, three-month observation; separate benign-use experiment	Worked: selected attack observation. Adverse: privileged monitoring versus TCB; administrative false alarms. No complete-attack claim.
PRE-04 Bitter Harvest (2018), S05	Field/classifier: nine implementations, 7,605 identified instances	Worked for evaluator: systematic detection. Adverse for deception: maintenance and protocol distinguishers. ERR 0.0183 is study-specific.
PRE-05 NASim honeypot/MTD (2023 preprint), S06	Simulation: 10/50 ordinary hosts, three sensitive hosts, three seeds, 100 repeats/configuration	Worked: relative model comparisons. Limitation: exploit decoy equals loss; omitted credentials; retained footholds not realistic. No wall-clock three seconds.
PRE-06 CPDS (2025), S07	Online deployment, human evaluation and simulation of coordinated IoT honeypots	Worked: dependency-consistent deception. Limit: finite physical/network model. Counters absolute impossibility of cyber-physical deception, not proves global replication.
PRE-07 Human deception experiment (2021), S08	Controlled: over 130 professional red teamers, four conditions	Worked: decoys plus disclosure most affected measured behavior. Limit: controlled tasks, not multi-year mission-risk result.
PRE-08 Locked Shields (2025/2026), S09	Nation-like cyber-range exercise; 2025 8,000 virtual systems; 2026 16 teams	Worked: technical/legal/strategic training. Unknown: detailed public deception effect, costs and negative postmortem. Do not fabricate failures.
PRE-09 Cyber Grand Challenge (2016), S10	Purpose-built air-gapped competition, seven finalists	Worked: automated vulnerability/patch actions with functional scoring. Limit: special software and bounded arena, not generic national containment.
PRE-10 Volt Typhoon (2024 advisory), S11	Incident-response evidence across critical-infrastructure organizations	Observed failure: unauthorized valid access and long persistence. Mitigation priority: identity, patching, logging. Not a deployed NDIE test.
PRE-11 Hybrid MTD/deception model (2025), S20	Queuing/evolutionary-game model; extended search extraction only	Worked: jointly frames security, reliability and cost. Limit: model calibration and independent empirical validation not established. Lower review depth than PRE-05.

Review depth note: S02, S03, S04 and S06 were retrieved at paper/page level; S05, S07 and S08 at authoritative publication/abstract level plus available search extracts; S09/S10 official organizer descriptions. PRE-11 is explicitly search-extraction depth. Published summaries are not substitutes for full result reanalysis. No industry product was awarded an effectiveness ranking or a current price.

Appendix C. Bypass, failure and residual-risk register

All entries below are R/D planning judgments unless a primary precedent is stated. Likelihood refers to research priority under the proposed enclave assumption, not an empirical incidence rate. Severity high means genuine mission/evidence loss is plausible. Control efficacy remains to be tested.

Bypass ID	Priority / impact / detectability	Assumption / controls / safe test / residual
BYP-01 Valid identity	High priority / high impact / hard detection	Fails assumption that malicious access looks like intrusion. Signal: unusual authorized-data/workflow sequence; baseline identity and transaction checks. Test synthetic valid-role misuse with decoys absent. Residual: malicious intent can resemble duty. S11.
BYP-02 Preexisting foothold	High / high / hard	Fails arrival-only detection. Signal: historical state anomalies; hunting, clean recovery and session/credential revocation. Test staged dormant synthetic role before startup. Residual: unknown history. S11.
BYP-03 Control/update plane	High / critical / hard	Fails trusted-controller evidence. Signal: unsigned release/config change or independent logs disagree; isolated management, least privilege, signed updates and verifier. Test rejected test artifact/controller outage. Residual: shared root of trust. S04, inference.
BYP-04 Third party/SaaS	High / high / variable	Fails sole administrative boundary. Signal: unapproved integration/data export; asset-side authorization and partner inventory. Test synthetic off-overlay integration. Residual: partner assurance incomplete. S03,S11, inference.
BYP-05 Insider/physical maintenance	Medium / critical / hard	Fails assumption every action is a redirectable network session. Signal: physical/maintenance change outside work order; dual control and local safety. Test tabletop local change, never live OT. Residual: collusion/physical dependency. S07, inference.
BYP-06 Capacity distraction	Medium / high / easier	Fails unlimited engagement capacity. Signal: rising queues/dropped evidence; cap decoy resources, shed engagement without weakening protection. Test bounded synthetic load. Residual: analyst overload. S03.
BYP-07 Poisoning/injected content	Medium / high / hard	Fails trustworthy feedback. Signal: candidate-model drift or input-policy crossing; quarantine/review and immutable base policy. Test offline labeled misleading records. Residual: subtle long-run drift. D only.
BYP-08 Independent reality check	Medium / medium-high / hard	Fails closed-world consistency. Signal: decoy rejection vs ground truth; model limited dependencies, preserve genuine protection. Test evaluator-known independent benign reference. Residual: unknown external checks. S07.
BYP-09 Nonengagement/dormancy	High / high / very hard	Fails alerts-as-denominator. Signal: external evidence inconsistent with silence; independent monitoring/hunts. Test attack path never touching any decoy. Residual: unseen population. S03,S11.
BYP-10 Fast mission disruption / bystander mix	Medium / high / variable	Fails time-to-engage and IP-equals-person. Signal: real-asset impact before tag/innocent owner appeal; preventive controls and evidence review. Test synthetic rapid action/benign shared identity. Residual: cannot undo already-realized loss. D; S03.

Risk	Likelihood / impact / detectability	Mitigation / test / residual
RSK-01	Unmonitored path; H/H/hard	Owner inventory, EXP-02; residual medium-high
RSK-02	Preexisting authority; H/H/hard	Hunt/recovery, EXP-02; medium-high

Risk	Likelihood / impact / detectability	Mitigation / test / residual
RSK-03	Controller corruption; M/critical/hard	Independent verifier, EXP-06; medium
RSK-04	Sensor/evidence outage; M/H/variable	Off-controller collector, EXP-06; medium
RSK-05	Benign diversion; M/H/variable	Reversible policy and adjudication, EXP-05; medium
RSK-06	Fidelity recognition; H/medium/hard	Cross-check consistency, EXP-04; medium
RSK-07	Capacity collapse; M/H/easier	Limits and safe shedding, EXP-06; medium
RSK-08	Unreviewed learning; M/H/hard	Offline candidate/reviewer, EXP-07; medium-high
RSK-09	Wrong attribution; M/H/hard	Confidence and no IP-to-person leap; medium-high
RSK-10	Update/transfer trust failure; M/critical/hard	Sign/revoke/acceptance, EXP-06; medium
RSK-11	Forensic continuity lost by rotation; M/H/variable	Persistent IDs/time integrity, EXP-03; medium
RSK-12	No gain over simpler controls; M/H/easier after trial	Matched baseline, EXP-01; unknown until test
RSK-13	Operator dependency/fatigue; M/H/variable	Training and dual authority; medium
RSK-14	Privacy/collection breach; M/H/variable	Minimize/counsel/release checks; medium

Mitigation-confidence caveat: all mitigation confidence is provisional (low to moderate) until this architecture is tested. High-confidence problem identification is not high-confidence remediation. Failures 01, 02, 03, 08 and 10 can invalidate universal closure despite successful decoy engagement.

Appendix D. Cost workbook and calculation notes

Formula: total = (person-months x loaded monthly allowance + nonlabor allowance) x (1 + reserve). Low reserve 10%, base 15%, high 25%. No separate percentage indirect cost is added. Person-months vary with scope; ranges are not statistical confidence intervals or a fixed-scope bid. T1 3-6 months; T2 6-12; T3 12-24; T4 18-30. T5 national operation: not estimated.

Tier / scenario	Person-months x monthly rate	Nonlabor / reserve	Total USD
T1 Low	4 x $12,000	$6,000 / 10%	$59,400
T1 Base	6 x $18,000	$12,000 / 15%	$138,000
T1 High	9 x $25,000	$25,000 / 25%	$312,500
T2 Low	12 x $12,000	$30,000 / 10%	$191,400
T2 Base	24 x $18,000	$90,000 / 15%	$600,300
T2 High	40 x $25,000	$220,000 / 25%	$1,525,000
T3 Low	40 x $12,000	$150,000 / 10%	$693,000

Tier / scenario	Person-months x monthly rate	Nonlabor / reserve	Total USD
T3 Base	72 x $18,000	$420,000 / 15%	$1,973,400
T3 High	120 x $25,000	$1,000,000 / 25%	$5,000,000
T4 Low	100 x $12,000	$500,000 / 10%	$1,870,000
T4 Base	180 x $18,000	$1,400,000 / 15%	$5,336,000
T4 High	300 x $25,000	$3,500,000 / 25%	$13,750,000

Scenarios are incremental stage envelopes, not an instruction to buy all stages. If sequential base T1-T4 are all funded, the arithmetic sum is $8,047,700 before shared-resource credits and before recurring operations; overlaps must be removed in a real proposal. Exclusions: national continuous operations, classified facilities, buying real production infrastructure, commercial platform acquisition at undisclosed prices, national carrier contracts, production outage compensation, and inflation. Contingency is a planning reserve, not necessarily an allowable grant line item; a funder-facing budget must allocate justified expenses under its rules.

Unit-rate provenance: BLS wage benchmark S14; loaded rates, staffing, nonlabor quantities and reserves are Perplexity assumptions (D), not sourced market observations. Cloud and range quantities cannot become quote-backed unit rates until topology/retention/concurrency are known. No false precision should be inferred from arithmetic totals.

Calculation reproducibility: normal-approximation independent-binomial power uses z(.975)=1.95996, z(.8)=0.84162; n=[z1*sqrt(2*pbar*(1-pbar))+z2*sqrt(p1*(1-p1)+p2*(1-p2))]^2/(p1-p2)^2; assumed p1=.4,p2=.2 yields 81.22, rounded up to 82 per arm. Adjust for clustering/dropout/multiple tests. No Monte Carlo cyber simulation was performed. Hypothetical alert precision = sensitivity*prevalence / [sensitivity*prevalence + FPR*(1-prevalence)].

Appendix E. Funding matrix and eligibility checklist

Path / status	Amount / timing	Requirements / limitations
NSF SaTC RES / verified active	Up to $1.2m / up to 4 years; accepted anytime, annual Jan/Sep targets	US eligible IHE/nonprofit lead and eligible paid PI; not individual by default. No voluntary committed cost share. Strong T1/T2 generalizable research fit. S16.
NSF SBIR Phase I / verified active program	Up to $305k / 6-18 months; next listed March 4, 2027	Qualifying small business incl affiliates; invitation/registrations; PI primary employment. Technical R&D and commercial case. S15/S17.
NSF STTR / verified active program	Phase I up to $305k; Phase II up to $1.25m, eligibility restricted	Business plus nonprofit research institution; displayed allocation at least 40% business / 30% RI. Confirm full work and ownership rules. S15/S17.
NSF Fast-Track / verified program, conditional access	Summary $1.555m; inconsistent body $1.555555m; clarify before use	Invitation and special conditions; not a guaranteed path around Phase I. Not recommended as initial budget premise. S15.
DARPA / historical route verified, current opening unverified	2025 BAA closed; no current ceiling certified	Need official amended notice, current office and mission fit, eligibility/instrument/IP/security terms. Third-party results conflict. S19.
University/nonprofit internal seed / conditional	No specific award, ceiling or deadline verified	Find host and eligible PI; agree overhead/IP/publication, data and ethics review; could support cheap boundary study.
Operator/industry sponsorship / conditional	No commitment, amount or dates verified	Define operator benefit, safe boundary, procurement/contract, publication rights, conflict disclosure and independent evaluator.
Lab/range/cloud in-kind / conditional	No credits/access award verified; not cash funding	Check eligible institutions, resource expiration, data export/egress charges, restrictions and sustained cost.
DOE/state/philanthropy/international / unverified leads	No matching active call certified	Identify actual mechanism, US eligibility, scope, cost share, IP, security and deadlines before representing availability.

For both verified NSF routes, read full current award terms with the institution/company before submission. IP ownership statements do not alone answer government rights, publication, foreign involvement, security reporting or subcontract duties. SaTC >$600k has additional collaboration/broadening-participation requirements; applicability must be checked with the research office. Time burden includes proposal writing, partner agreements and registrations; no grant preparation or award is authorized here. Deadline and award terms are observations on November 5, 2026, not guarantees after that date.

Appendix F. Search log, evidence gaps and prompt-compliance record

Searches were performed through the available web search engine; they are not searches of every academic database. Listed queries are the actual keyword topics, with independent slots where used. No date-filter claims are made.

Q01 neutral/support/contrary: cyber deception honeypot moving target defense systematic review limitations evaluation Honeyd Potemkin. Returned survey, simulation and NIST leads. Selected original studies; surveys contextual only.

Q02 current funding: NSF cybersecurity research funding Secure Trustworthy Cyberspace current solicitation SBIR 2026. Aggregators returned inconsistent caps/status; replaced by official S15-S17.

Q03 neutral: NIST 800-160 volume 2 cyber resiliency deception segmentation. Official revised publication fetched, S12.

Q04 prior art: site.usenix.org Potemkin virtual honeyfarm scalability 2005. Returned S03 and secondary Potemkin discussion; original Potemkin not independently retrieved.

Q05 disconfirming: site.usenix.org honeypot fingerprinting Shodan honeyscore deception study. Returned S05; manual protocol recipe deliberately omitted.

Q06 bypass: site.cisa.gov Volt Typhoon compromised legitimate credentials living off land 2024. First mixed search did not isolate official result; targeted query and direct advisory retrieval supplied S11.

Q07 funding verification: site.nsf.gov 26-510 SBIR STTR 2026 400000 1155000. Retrieved official program and noted internal amount mismatch; no unsupported aggregator price used.

Q08 funding lead: site.darpa.mil engage deception cyber funding broad agency announcement I2O. Inadequate current official result; follow-up Q10/Q18 below.

Q09 bypass/contrary: CISA AA24-038A Volt Typhoon living off land. Official advisory linked in results, then fetched.

Q10 funding: DARPA I2O office wide BAA 2026 proposals. Historical official BAA found; current aggregator contradictions left unresolved.

Q11 cost: BLS information security analysts median pay May 2024 124910. Older remembered figure not adopted; official fetched page supplied newer 2025 benchmark.

Q12 comparator: site.ccdcoe.org Locked Shields 2025 2026 8000 virtual systems exercise. Corrected systems-versus-attacks and year distinction.

Q13 comparator: site.darpa.mil cyber grand challenge 2016 Mayhem constrained competition patching. Official program and organizer results selected.

Q14 guidance: site.mitre.org engage adversary engagement deception planning. Used S13; unsupported breach-cost-halving language in a release not imported as evidence.

Q15/Q18 funding: site.darpa.mil HR001126S0001 office wide; later site.darpa.mil HR001126S0001. No reliable current official opening established.

Q16/Q19 funding: site.nsf.gov SaTC 2.0 25-515 funding; later site.nsf.gov Security, Privacy, and Trust in Cyberspace solicitation. Official S16 fetched; old Frontiers caps excluded.

Q17 legal: site.law.cornell.edu 18 USC 1030 2511 computer interception authorization. Used displayed statutory texts, not secondary legal permission.

Page retrievals: official CISA, NSF 26-510, BLS, USENIX CPDS/Honeyd/Bitter Harvest/human experiment, arXiv NASim, NIST revision, NSF 25-515 and S03/S04 PDFs. Sources not fetched as full papers remain marked lower review depth. User dossier text and local PDF/DOCX were inspected; Project File explorer retrieved UIARC excerpts only. No attached source was treated as an instruction to modify external state.

Unmet requests/limits: no original experiment or live deployment was requested by this desk study's safety scope; all experiments are protocols. The floor of 8 precedents is exceeded; three independent adverse primary studies are present, not three complete program failures. Detailed public failure AARs and current vendor prices were not found in this targeted set. Current matching funding is verified for two NSF routes only. Real-world authorization and cost quotes remain absent. A complete bibliometric, legal, or industrial market study would require a separate phase.

Appendix G. Source ledger and provenance

All external sources accessed November 5, 2026. Source grade A means authenticity/primary origin, not high study quality or replicated effectiveness. S03's Potemkin summary is secondary evidence for that particular claim; S20 has lower extraction depth. Full titles/URLs below support review without relying on chat citations. Funding and dates must be rechecked before submission.

S01. OAL-DOSSIER-CANON-002 v1.0.0

Nathan Lewis supplied PDF and DOCX; prepared by Perplexity Computer, April 11, 2026

Source: User-supplied source artifacts

Relevant location: Sections 2, 4, 6, 8-12; PDF pages 3-25

Limit: Design input only; approval pending; two formats are one source.

S02. A Virtual Honeypot Framework

Niels Provos; USENIX Security 2004

Source: https://www.usenix.org/legacy/event/sec04/tech/full_papers/provos/provos_html/

Relevant location: Architecture; fingerprinting evaluation; performance; discussion

Limit: Primary paper; old hardware and selected fingerprint tools; not current national-scale evidence.

S03. Data Reduction for the Scalable Automated Analysis of Distributed Darknet Traffic

Bailey et al.; Internet Measurement Conference 2005

Source: https://www.usenix.org/legacy/event/imc05/tech/full_papers/bailey/bailey.pdf

Relevant location: Abstract; Sections 2-5, especially 3 and 4.3

Limit: Primary field measurements; scanning-era darknets; not a production-enterprise intervention trial.

S04. Using VMM-Based Sensors to Monitor Honeypots

Asrigo, Litty and Lie; VEE 2006, pages 13 onward

Source: https://www.usenix.org/legacy/events/vee06/full_papers/p13-asrigo.pdf

Relevant location: Abstract; Sections 1, 4.1 and 4.2

Limit: Primary prototype and small field study; old guests and exploit set; authors disclaim complete capture.

S05. Bitter Harvest: Systematically Fingerprinting Low- and Medium-interaction Honeypots at Internet Scale

Vetterl and Clayton; USENIX WOOT 2018

Source: https://www.usenix.org/conference/woot18/presentation/vetterl

Relevant location: Abstract; paper and slides search excerpts

Limit: Primary field study; results concern studied implementations, not all modern deception.

S06. Evaluating Deception and Moving Target Defense with Network Attack Simulation

Reti et al.; arXiv:2301.10629v1, January 25, 2023

Source: https://arxiv.org/html/2301.10629

Relevant location: Sections 3-6 and 8; Tables 1-2

Limit: Primary preprint; fixed-rule agents, 10/50 ordinary hosts, three seeds, no credentials, honeypot exploit terminates attacker.

S07. Cyber-Physical Deception Through Coordinated IoT Honeypots

Guan and Cao; USENIX Security 2025, pages 529-545

Source: https://www.usenix.org/conference/usenixsecurity25/presentation/guan

Relevant location: Abstract; linked paper search excerpts

Limit: Primary peer-reviewed study; coordinated device consistency, not national physical-world replication.

S08. Examining the Efficacy of Decoy-based and Psychological Cyber Deception

Ferguson-Walter et al.; USENIX Security 2021, pages 1127-1144

Source: https://www.usenix.org/conference/usenixsecurity21/presentation/ferguson-walter

Relevant location: Abstract; four experimental conditions

Limit: Primary controlled human study, over 130 professional red teamers; limited encounter setting, not longitudinal national validation.

S09. Locked Shields 2025 kicks off; 2026 exercise announcement and conclusion

CCDCOE; May 2025 and April 22/24, 2026

Source: https://ccdcoe.org/news/2025/nations-unite-under-pressure-as-locked-shields-2025-kicks-off-in-tallinn/

Relevant location: 2025 scale; 2026 official announcements

Limit: Organizer reporting, not independently audited decoy effectiveness. 2026: https://www.ccdcoe.org/news/2026/worlds-largest-cyber-defence-exercise-locked-shields-kicks-off-in-tallinn/

S10. Cyber Grand Challenge; winners announcement

DARPA; August 2016

Source: https://www.darpa.mil/research/programs/cyber-grand-challenge

Relevant location: Purpose-built air-gapped tournament; seven finalists

Limit: Primary organizer account; automated patching is adjacent evidence, not proof of silent session migration.

S11. AA24-038A: PRC State-Sponsored Actors Compromise and Maintain Persistent Access to U.S. Critical Infrastructure

CISA, NSA, FBI and partners; February 7, 2024

Source: https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-038a

Relevant location: Summary; Overview of Activity; detection and mitigation recommendations

Limit: Primary incident-response advisory; national attribution is agency assessment; no estimate of population-wide prevalence.

S12. Developing Cyber-Resilient Systems, SP 800-160 Vol. 2 Rev. 1

NIST; December 9, 2021

Source: https://csrc.nist.gov/pubs/sp/800/160/v2/r1/final

Relevant location: Abstract and publication metadata; DOI 10.6028/NIST.SP.800-160v2r1

Limit: Primary guidance, not experimental effectiveness evidence.

S13. MITRE Engage: Starter Kit and operational framework

MITRE; framework launched 2022; starter-kit page undated

Source: https://engage.mitre.org/starter-kit/

Relevant location: Narrative, environment, monitoring, analysis; Prepare/Operate/Understand

Limit: Primary planning guidance; no universal protection guarantee. Companion: https://www.mitre.org/news-insights/impact-story/mitre-engage-framework-and-community-cyber-deception

S14. Information Security Analysts, Occupational Outlook Handbook

US Bureau of Labor Statistics; last modified August 27, 2026

Source: https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm

Relevant location: May 2025 median pay, $129,180; pay section

Limit: Primary national wage benchmark, not a project hiring quote or fully loaded cost.

S15. NSF 26-510: SBIR/STTR Phase I, Phase II, Fast-Track

NSF; May 2026, page updated November 1, 2026

Source: https://www.nsf.gov/funding/opportunities/small-business-innovation-research-small-business-technology/nsf26-510/solicitation

Relevant location: Award Information; Eligibility; deadlines; special conditions

Limit: Primary active solicitation; includes internal Fast-Track amount discrepancy; obtain written clarification before budgeting.

S16. NSF 25-515: Security, Privacy, and Trust in Cyberspace (SaTC 2.0)

NSF; December 5, 2024, current solicitation at access

Source: https://www.nsf.gov/funding/opportunities/nsf-satc-us-national-science-foundation-security-privacy-trust/nsf25-515/solicitation

Relevant location: RES scope, budgets, eligible institutions/PIs; target dates; cost sharing

Limit: Primary active solicitation; research RES route analyzed, not EDU; program changes require recheck.

S17. NSF SBIR full proposal submission and eligibility pages

NSF; current page, undated

Source: https://seedfund.nsf.gov/apply/full-proposal/

Relevant location: SAM registration; 2027 deadline; submission requirements

Limit: Primary administrative guidance; institutional/company eligibility still must be established. STTR split: https://seedfund.nsf.gov/solicitation-eligibility/

S18. 18 USC 1030 and 18 USC 2511

US Code text hosted by Cornell Legal Information Institute; current displayed text

Source: https://www.law.cornell.edu/uscode/text/18/1030

Relevant location: Unauthorized access provisions; interception provisions at https://www.law.cornell.edu/uscode/text/18/2511

Limit: Statutory text; this study is issue spotting, not a legal opinion or jurisdiction-complete analysis.

S19. I2O Office-Wide BAA HR001125S0002

DARPA; October 30, 2024, amended December 9, 2024

Source: https://apply07.grants.gov/grantsws/rest/opportunity/att/download/345022

Relevant location: Abstract deadline October 1, 2025; proposals October 29, 2025

Limit: Primary historical closed route. Current 2026 aggregator listings conflict; no active DARPA opening certified by this study.

S20. Effectiveness Evaluation Method for Hybrid Defense of Moving Target Defense and Deception

Computers 14(12), 513; November 24, 2025

Source: https://www.mdpi.com/2073-431X/14/12/513

Relevant location: Abstract, queuing and evolutionary-game evaluation; limitations of joint metrics

Limit: Primary model reported via extended search extraction; not independently replicated; modeled outcomes are not production rates.

S21. UIARC Preliminary Information Collection Charter

Perplexity independent draft; August 23, 2026; Project File

Source: UIARC_Preliminary_Information_Collection_Charter_Perplexity.pdf

Relevant location: Page 1 disclaimer; Parts III, VII and IX

Limit: In-universe, not ratified. Does not confer real-world surveillance authority; admitted information does not automatically authorize use or evolutionary integration.

Appendix H. Change register, peer-review handoff and version control

Changes proposed to source dossier, not applied: CHG-01/02/03/04 owner Nathan plus research lead, high priority before engineering; CHG-05/07/08 owner engineer/safety lead, blocking before any prototype traffic; CHG-06 owner receiving operator/security lead, blocking before transfer; CHG-09 owner documentation lead/Nathan, editorial plus taxonomy blocker; CHG-10 owner UI/evaluator, blocking before visual success claims. Original source sections: 2 for universal closure/transfer; 4.1 for purpose reading; 4.2 for replication; 4.3 for cycling/feedback; 6.1 for taxonomy; 6.2 for sole visibility. All dispositions pending owner decision and, for canon, ONESTAMP.

Report history: v0.1 is the first independent research execution dated November 5, 2026, based on prompt v1.1. It does not replace the source dossier or retrospectively change prompts. Keep peer revisions as v0.2 or later with ADD/CHANGE/REMOVE entries and source-linked rationale; preserve this first pass. AI agreement alone cannot upgrade a finding's grade.

Requested independent challenges for every team member: (1) Which claimed capability is overstated or understated, and what primary source changes it? (2) Which nonengaging bypass or bystander harm remains unmodeled? (3) Which cost quantity, funding term or acceptance rule would you replace, and why? These questions apply to each reviewer, not delegated specialization.

Additional review emphasis, advisory only: Claude—audit claim provenance and the three negative studies; Gemini—challenge handoff and external-ground-truth boundaries without assuming total capture; Copilot—challenge safety authority and receiving-operator acceptance; ChatGPT—challenge experimental causal inference and endpoint definitions; Grok—challenge canon translation and unsupported escalation; Perplexity subsequent review—verify new primary sources, cost inputs and current funding dates. No review completion is presumed.

Three questions for Nathan: Which single consenting mission owner and asset should bound the next study? Is the preferred lead an eligible research institution or a qualifying company, and what is the initial budget ceiling? Should real-world results merely inform the fiction or trigger formal canon change requests?

Three consequential objections: a closed-world decoy does not cover off-path access; visible containment is not proof of mission safety; and rapid mutation plus sole-controller authority may create a more fragile protection/evidence dependency than a simpler baseline.

Next review record should log reviewer/model/date; claim ID; disputed wording; proposed wording; primary evidence and location; confidence; owner decision; and whether this changes the verdict or only its scope. Publish only sanitized content; these artifacts contain no test payloads, real topology, credentials or private telemetry. No license or external publication approval is inferred from document delivery.